Problem 9: Super Ben Galaxy 2
11 Points

Problem ID: `galaxy2`
Rank: `3`

Introduction

Now, let me tell you a new story that also begins with paint...

Shimmering paint falls on UC Berkeley once a semester. That time had come once again...

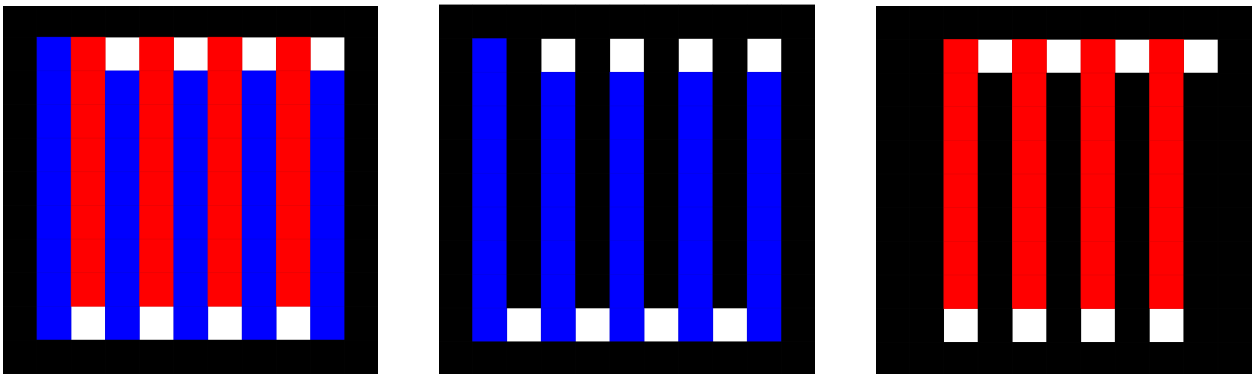


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Problem Statement

This is the second of a two part problem series! You can find the first part at `galaxy`. Key differences introduced in this second part are highlighted for emphasis.

You are given an $\mathbf{N} \times \mathbf{M}$ grid $\mathbf{G}$, consisting of red (`R`), blue (`B`) and normal (`.`) tiles. You are only able to stand or walk onto tiles which are active. Normal tiles are always active, red tiles are active only in red tile mode, and blue tiles are active only in blue tile mode. If you are standing on a normal tile, you may choose to perform a spin action, which toggles which mode you are in (red tile mode to blue tile mode or vice versa). See Figure 1 below for a visualization of the third sample test case, with white representing normal tiles and black representing tiles that cannot be walked on:



(a) Full Map (b) Blue Tile Mode (c) Red Tile Mode

Figure 1: Visualizations of Sample Test Case 3

You are given $\mathbf{Q}$ queries, each of which contains Ben's starting location at $(\mathbf{R_{start}}[i], \mathbf{C_{start}}[i])$, Ben's intended destination at $(\mathbf{R_{end}}[i], \mathbf{C_{end}}[i])$, and the time limit $\mathbf{K}$ in seconds up to which Ben has to get from start to end. Moving to an adjacent tile takes exactly 1 second, and each spin action takes exactly 1 second. You may assume that the tile at Ben's starting location is normal, and that we start in blue tile mode.

For each query, return whether Ben can successfully get from his starting location at $(\mathbf{R_{start}}[i], \mathbf{C_{start}}[i])$ to his destination at $(\mathbf{R_{end}}[i], \mathbf{C_{end}}[i])$ within the time limit of $\mathbf{K}$ seconds.

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Input Format

The first line of the input contains a single integer $\mathbf{T}$ denoting the number of test cases that follow. For each test case:

- The first line contains three space-separated integers $\mathbf{N}$ $\mathbf{M}$ $\mathbf{Q}$, where $\mathbf{N}$ is the number of rows, $\mathbf{M}$ is the number of columns, and $\mathbf{Q}$ is the number of queries.
- Each of the next $\mathbf{N}$ lines contains a string of length $\mathbf{M}$ representing a row in the grid. Each character in the string is either `R`, `B` or `.`
- Each of the next $\mathbf{Q}$ lines is a query, the ith of which contains space-separated integers $\mathbf{R_{start}}[i]$ $\mathbf{C_{start}}[i]$ $\mathbf{R_{end}}[i]$ $\mathbf{C_{end}}[i]$ $\mathbf{K}$ in that order, representing the zero-indexed coordinates of Ben's starting location and destination, and then the time limit to get from start to end.

Output Format

For each test case, output $\mathbf{Q}$ lines. Each line should either be `YES` if it is possible for Ben to go from his starting location to his destination within the time limit of $\mathbf{K}$ seconds for the corresponding query, or `NO` if not.

Constraints

Time Limit: **4 seconds**

$1 \leq \mathbf{T} \leq 100$
$1 \leq \mathbf{N} \leq 10$
$1 \leq \mathbf{M} \leq 4000$
$1 \leq \mathbf{Q} \leq 10^5$
$0 \leq \mathbf{R_{start}}[i], \mathbf{R_{end}}[i] < \mathbf{N}$ for all i
$0 \leq \mathbf{C_{start}}[i], \mathbf{C_{end}}[i] < \mathbf{M}$ for all i
$0 \leq \mathbf{K} \leq 10^9$
The sum of $\mathbf{N} \times \mathbf{M}$ over all test cases is at most 4×10^4.
The sum of $\mathbf{Q}$ over all test cases is at most 10^5.

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Sample Test Cases

Sample Input Download

```
3
2 4 2
.R.B
.RRB
0 0 1 3 6
1 0 1 3 6
2 4 2
.RB.
.RB.
1 0 1 2 10
1 3 0 3 1
9 9 3
BR.R.R.R.
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
B.B.B.B.B
0 8 4 5 8
8 1 2 8 12
8 7 0 1 20
```

Sample Output Download

```
YES
NO
NO
YES
YES
NO
YES
```

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Sample Explanations

For the following diagrams, green tiles represent the current tile that Ben is standing on.

Figure 2 shows the full map for the first sample test case, with both blue and red tiles active for the purposes of visualization. Note that in the actual problem, blue and red tiles will never be active at the same time.

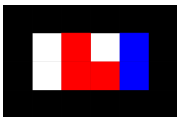


Figure 2: Full Map

For the first test case's first query, Figure 3 displays the initial state of the grid in blue mode, and Ben is at his starting location in the top left corner.

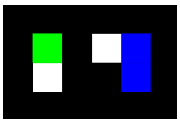


Figure 3: Initial State

Ben then performs a spin action while at the top left corner, taking 1 second and switching the state of the grid to red mode in Figure 4, and Ben is able to move to the cell at $(0, 2)$ by moving right 2 steps, taking 2 seconds.

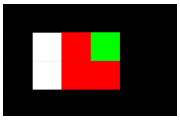


Figure 4: First Spin

Ben uses a second spin action at this cell, taking 1 second to switch the grid back to blue mode, and he is able to reach the bottom right corner in 2 seconds by taking 1 step right and 1 step down, as shown in Figure 5.

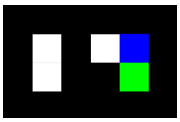


Figure 5: Second Spin

In total, Ben took $1 + 2 + 1 + 2 = 6$ seconds to get to his destination, which is within the time limit of 6 seconds for this query.

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